Optimization Problems.

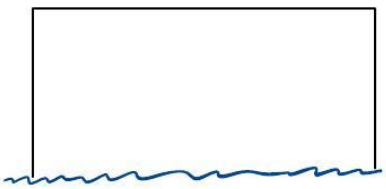
Here we will study applications of the first and second derivative test that is, using maximum and minimum.
The methods we have learned in max/min section for finding extreme values which have many applications in the real life.

Steps in Solving Optimization Problems:

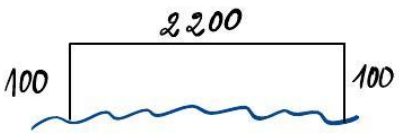
1. **Understand the problem:** The first step is to read the problem carefully until it's clearly understood.
2. **Draw a Diagram.**
3. **Introduce Notation (Using new variables).**
4. **Use these variables and obtain a relationship between them.**
5. **Obtain a function in only one variable.**

Example 1. A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field tha has the largest are?

Sln: step 2.

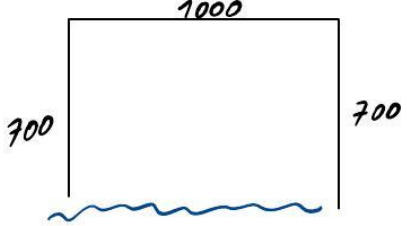


Notice that: ⚠



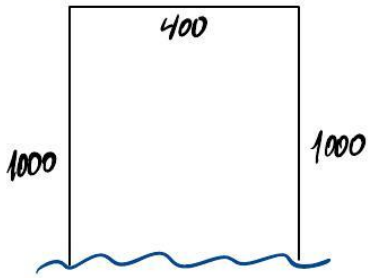
$$A = (100)(2200) = 220{,}000\ ft^2$$

$$P = 100 + 2200 + 100 = 2400\ ft \checkmark$$



$A = (700)(1000) = 700{,}000\ ft^2$

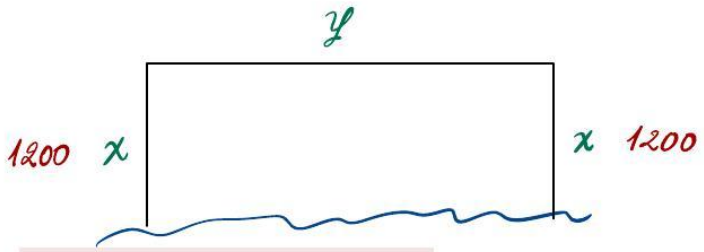
$P = 700 + 1000 + 700 = 2400\ ft$ ✓



$A = (1000)(400) = 400{,}000\ ft^2$

$P = 1000 + 400 + 1000 = 2400$ ✓

Step 3.



Restriction: $x + y + x = 2400$

We are looking for the largest area:

my "function" is the area function:

$$A = x \cdot y$$

We have two variables.

Using my restriction: $x + y + x = 2400$

$$2x + y = 2400$$

$$y = 2400 - 2x.$$

then

$$A(x) = x \cdot (2400 - 2x).$$

$$A(x) = 2400x - 2x^2, \quad 0 \le x \le 1200.$$

Thm from 224: If f is a continuous function on a closed interval $I = [a, b]$ then f has a abs. max and abs. minimum.

Rmk. If f is defined on $(I=[a,b])$ a closed interval, then the critical points are points s.t $f'(x_0)=0$ and $f'(x_0)=$ DNE and $x_0=a$, $x_0=b$.

Solving the problem: $A(x)=2400x-2x^2$; $0\le x\le 1200$.

$A'(x)=2400-4x$. So, $2400-4x=0 \iff x=\frac{2400}{4}=600$.

is 600 the only critical point? No, b/c the function $A(x)$ is defined on a closed interval.

The critical points: $x=0$, $x=600$, $x=1200$

at $x=0 \Rightarrow A(0)=0(2400-2(0))=0$.

at $x=600 \Rightarrow A(600)=600(2400-2(600))=600(2400-1200)=600\cdot 1200$

$=720{,}000\ ft^2$

at $x=1200 \Rightarrow A(1200)=1200(2400-2(1200))=0$.

So, the maximum area is $720{,}000\ ft^2$ when $x=600\ ft$.

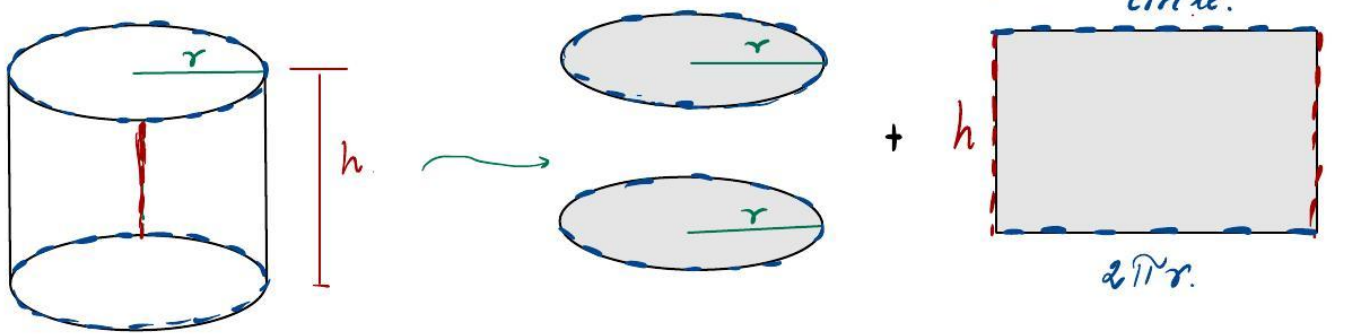
We need to find "y"

$y=2400-2(600)=1200\ ft$.

So, the rectangular field should be 600 ft deep and 1200 ft wide.

Example 2. A cylindrical can is to be made to hold 1 L of oil find the dimensions that will minimize the cost of the metal to manufacture the can.

Sln.



Then we are looking for the minimum of the area:

$$A = 2\pi r^2 + (2\pi r)h = 2\pi r^2 + 2\pi r h.$$

We have restriction that the volume is $1L = 1000\ cm^3$.

$$V = \pi r^2 \cdot h = 1000.$$

then solving for h:

$$h = \frac{1000}{\pi r^2}$$

$$A = 2\pi r^2 + 2\pi r \left(\frac{1000}{\pi r^2}\right)$$

$$A(r) = 2\pi r^2 + \frac{2000}{r}$$

then solving for r:

$$r = \sqrt{\frac{1000}{h}}$$

$$A = 2\pi\left(\sqrt{\frac{1000}{h}}\right)^2 + 2\pi\sqrt{\frac{1000}{h}} \cdot h$$

$$A(h) = \frac{2000\pi}{h} + 2\pi\sqrt{1000}\cdot\sqrt{h}.$$

Solving:

$$A'(r) = 4\pi r - \frac{2000}{r^2} = \frac{4\pi r^3 - 2000}{r^2}.$$

the only critical point is

$$r = \sqrt[3]{\frac{2000}{4\pi}} = \sqrt[3]{\frac{500}{\pi}}.$$

$$A''(r) = 4\pi + \frac{4000}{r^3}\ ;\ A''\left(\sqrt[3]{\frac{500}{\pi}}\right) = 4\pi + \frac{4000}{\left(\sqrt[3]{\frac{500}{\pi}}\right)^3} = 4\pi + \frac{4000}{500}\pi > 0$$

So, by the second derivative test we have a minimum at $r = \sqrt[3]{\frac{500}{\pi}}$.

and $h = \frac{1000}{\pi\left(\frac{500}{\pi}\right)^{2/3}}$.